

MapMyRun Field Quality Assessment

A balanced, evidence-bounded review of mobile resilience, workout-record integrity, user trust, and the next engineering questions.

Three obstruction scenarios produced plausible saved workouts. Separately, two predictable unended-session mistakes produced authoritative-looking Run data containing known vehicle travel. The app may have behaved exactly as designed.

Assessment type

External black-box field investigation

Primary quality question

Does a foreseeable human mistake merit a better prevention, detection, or recovery path?

Confidence

High in observations · medium in product interpretation ·
low in component attribution

Evidence boundary

No source, logs, raw GPS samples, requirements,
telemetry, or internal product history

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01 - Scope and method

This assessment evaluated user-visible MapMyRun behavior on an iPhone and Apple Watch using field notes, saved workout summaries, route maps, split tables, and analysis screens. It began with ordinary use and no defect hypothesis.

Two independent quality questions

- 1) When a Run remains active during vehicle travel, is the saved record trustworthy, explainable, and recoverable?
- 2) Under repeated GPS obstruction and different phone/Watch acquisition paths, does recording degrade gracefully?

Test matrix

ID	Activity / acquisition	Physical condition	Purpose
GC-01	Walk; iPhone-started; phone carried; Watch worn	Gold Camp Road; Tunnel #1 twice	Phone acquisition under obstruction
GC-02	Walk; Watch-started; powered-on phone left at start	Same practical course; tunnel twice	Watch acquisition and later sync
GC-03	Walk; Watch-started; phone powered off at start	Same practical course; tunnel twice	Remove paired-phone availability
SAT-WM	Run remained active	Pedestrian activity plus known driving	Accidental observation
SUN-WM	Run; Auto-Pause on; Watch worn	Walk, drive, stop, shop, drive, walk	Focused follow-up

Evidence rules

- Saved-workout and field timestamps were preserved separately.
- Road-following geometry supported location plausibility, not raw-sample accuracy.
- Gold Camp and vehicle-transition tracks were evaluated separately.
- No undocumented requirement was invented to make an observation look like a defect.
- AI interpretations were treated as hypotheses and challenged against the evidence.

02 - Evidence

Gold Camp resilience

All three walks traversed Tunnel #1 twice. Aggregate pace, heart rate, cadence, and elevation gain remained tightly grouped. Every workout saved and remained physically plausible.

Metric	GC-01	GC-02	GC-03
Duration	18:25	20:15	21:34
Distance	0.91 mi	1.00 mi	1.07 mi
Avg pace	20:07	20:18	20:05
Avg HR	125	124	124
Cadence	94	92	94

Supported: respectable reliability and graceful degradation under tested conditions.

Vehicle-transition records

Two separate Run records included known automobile travel and vehicle-scale pace.

Evidence	Saturday	Sunday
Saved summary	5.64 mi · 1:18:12	5.25 mi · 29:21
Vehicle-scale pace	2:02 mile	2:16–2:20 miles; 1:30 max
Avg HR / cadence	85 / 82	100 / 94
State handling	Not reconstructed	Auto-Pause on; paused observed

Supported: known vehicle movement entered authoritative-looking Run metrics. Root cause and product intent remain unverified.

Red-team correction: The 0.91–1.07 mile Gold Camp spread is not distance-defect evidence by itself. Durations increased while pace remained stable, and exact execution length was not controlled tightly enough to isolate accuracy.

Strongest cross-metric example

Sunday mile 2 saved as 2:16/mi while the split table reported HR 84 bpm and cadence 98 spm. That combination is not a credible description of human running, although the provenance and intended semantics of each sensor value remain unverified.

03 · Finding and product-quality question

“Working as designed” answers a requirements question. It does not automatically settle the product-quality question.

MapMyRun may have behaved exactly as designed. The quality question is whether a predictable human mistake deserves better prevention, detection, or recovery—balanced against false positives, accessibility, privacy, battery use, and implementation cost.

Why this matters

Workout records can feed history, trends, goals, coaching interpretation, social sharing, and downstream platforms. A technically complete record can still be semantically misleading. Precise values make the affected portion difficult for a user to recognize or repair.

Downstream user-trust signal

Apple Health later displayed the affected activity in a trusted daily view. This supports the user-trust concern; it does not establish exact synchronization timing, transformation behavior, responsible ownership, or broader downstream impact.

What was not concluded

- No confirmed implementation defect.
- No responsible component or algorithm.
- No fleet frequency or production severity.
- No claim that Outside has not already considered the behavior.
- No complete recovery assessment.
- No exact Apple Health propagation model.
- No optimal guardrail or threshold.
- No causal link from Gold Camp to the vehicle records.

Priority call: Do not assign severity from this sample alone. Use prevalence, support, recovery, and propagation data to set priority.

04 · Competing explanations and limitations

Hypothesis	Status	Rationale
Gold Camp GPS-obstruction resilience	Positive result	Six tunnel passages produced transient noise but three plausible saved walks.
Acquisition-path resilience	Positive signal	Phone-started, Watch-started, and phone-off scenarios produced similar plausible aggregates.
Active Run accepts vehicle movement	Observed twice	Known driving aligns with road routes and vehicle-scale splits.
Correct-by-design recording	Plausible	The system may intentionally record valid coordinates until the user stops it.
Raw GPS random or grossly displaced	Less consistent	Routes followed recognizable roads, but raw samples were not inspected.
Integrity signaling may be insufficient	Open product question	Selected Run plus 26–40 mph derived pace saved without an observed signal.
Watch, phone, or backend is responsible	Unknown	Attribution requires logs, provenance, and raw time series.

Known evidence gaps

- No raw timestamped location, speed, or accuracy series.
- No internal requirement for anomaly handling.
- No complete edit, trim, relabel, or recovery assessment.
- Only one user/account/device ecosystem.
- Apple Health showed a downstream trust signal, but exact sync timing and broader effects remain unverified.
- Field routes were similar, not laboratory-controlled repetitions.

These gaps limit causal and population claims. They do not erase the positive resilience result or the observed vehicle-scale Run metrics.

05 · Engineering strategy

1. Establish context before proposing a fix

Review requirements, support signals, prior product decisions, recovery behavior, and downstream ownership. Measure prevalence and impact before treating a field observation as a release blocker.

2. Reproduce deterministically

Capture or replay known-good pedestrian traces, then inject controlled vehicle segments, stops, poor-accuracy points, tunnels, stale fixes, and offline synchronization. Compare raw samples, accepted segments, aggregates, UI, sync, and downstream consumers.

3. Compare guardrails as product choices

Option	Potential benefit	Tradeoffs to test
Prevention / auto-stop	Stops contamination early	False positives; accessibility; unusual athletes; battery
Detection / confidence signal	Preserves data while communicating uncertainty	Explainability; alert fatigue; threshold complexity
Post-workout repair	Lets users trim, relabel, or confirm	Discoverability; user effort; downstream correction
No behavior change	Avoids new product and operating cost	Trust, support, and downstream-data consequences

4. Add regression properties and observability

- internal consistency
- explainability
- data preservation
- idempotent sync
- uncertainty
- privacy-aware decisions

Instrument why segments are accepted, rejected, or flagged; monitor outcomes by device, OS, app version, acquisition path, and workflow. Preserve Gold Camp as a positive resilience baseline.

06 · Management view and retrospective

Dimension	Assessment	Management response
Reliability	All three Gold Camp walks saved and remained plausible.	Preserve as a positive baseline.
User trust	Two unended Runs presented vehicle-scale performance as authoritative data.	Establish prevalence, recovery, and downstream impact.
Product policy	Documented behavior may still leave a foreseeable mistake unguarded.	Evaluate options and tradeoffs with product context.
Observability	External evidence cannot attribute sample acceptance or ownership.	Add privacy-aware decision telemetry and traceability.
Testability	Field testing found value but is not deterministic.	Create replayable traces, properties, and regression scenarios.

Field-test retrospective

The three Gold Camp paths were similar but not controlled tightly enough to attribute small distance differences to MapMyRun. A stronger next cycle would synchronize clocks, mark exact start/turn/end points, predefine variables and acceptance criteria, repeat each condition, separate execution from logging, and preserve the highest-resolution evidence available.

Final assessment

The strongest result is balanced: MapMyRun demonstrated respectable resilience under GPS obstruction, while a separate, predictable unended-session mistake exposed a legitimate user-trust question. The evidence supports further investigation—not a declaration that the product is broken.

Why this matters

Field science is iterative. The useful outcome is not pretending the first interpretation was perfect; it is showing where inference became weaker, preserving uncertainty, and improving the next experiment. That inspect-and-adapt loop is quality engineering.

AI assistance

AI accelerated research, test-design iteration, evidence organization, challenge of competing interpretations, visual production, and drafting. Michael Jensen executed the field tests, captured the observations, corrected unsupported connections, and retained ownership of the final judgments.